

02464

Artificial Intelligence and Human Cognition

Visual Perception – Neural encoding

Cognition as computing

Marr's three levels

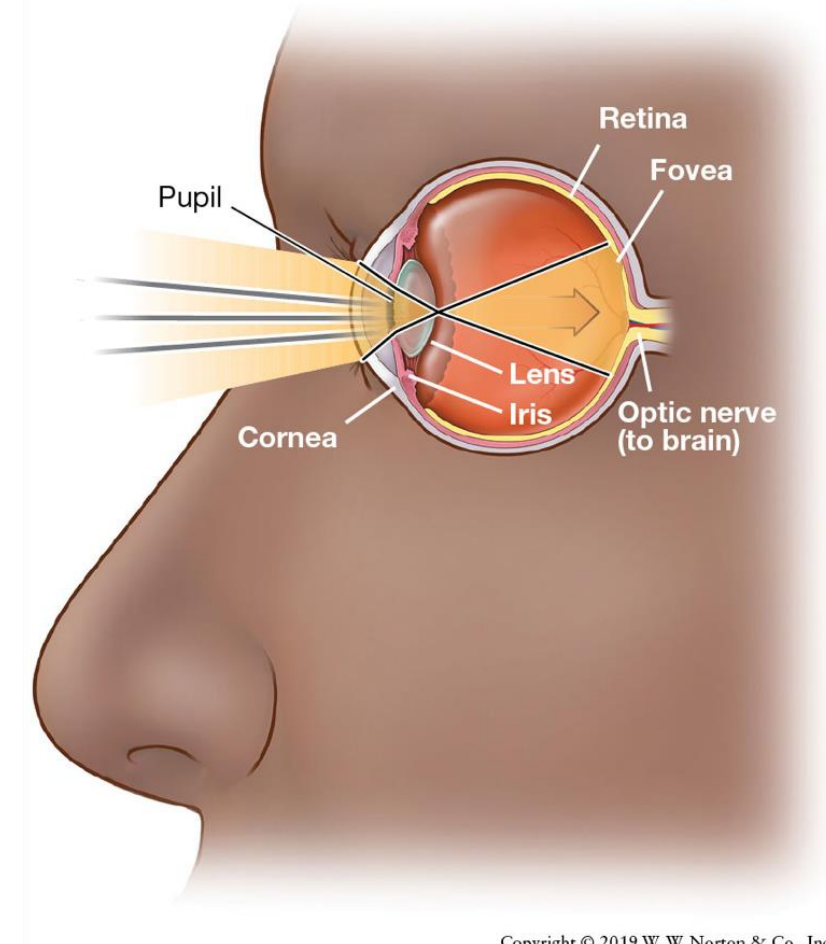
Implementational level: the physical realization of the system

- Defines the computer hardware
- Compare computers and the brain at this level
 - Computer hardware is known by design
 - Brain functions are uncovered by experimentation
 - Require measurement of brain activity
- Example: no dedicated graphics card, 4 GB RAM memory

Organization of the Eye

- Light pass through the cornea and lense
- Photoreceptors on the retina

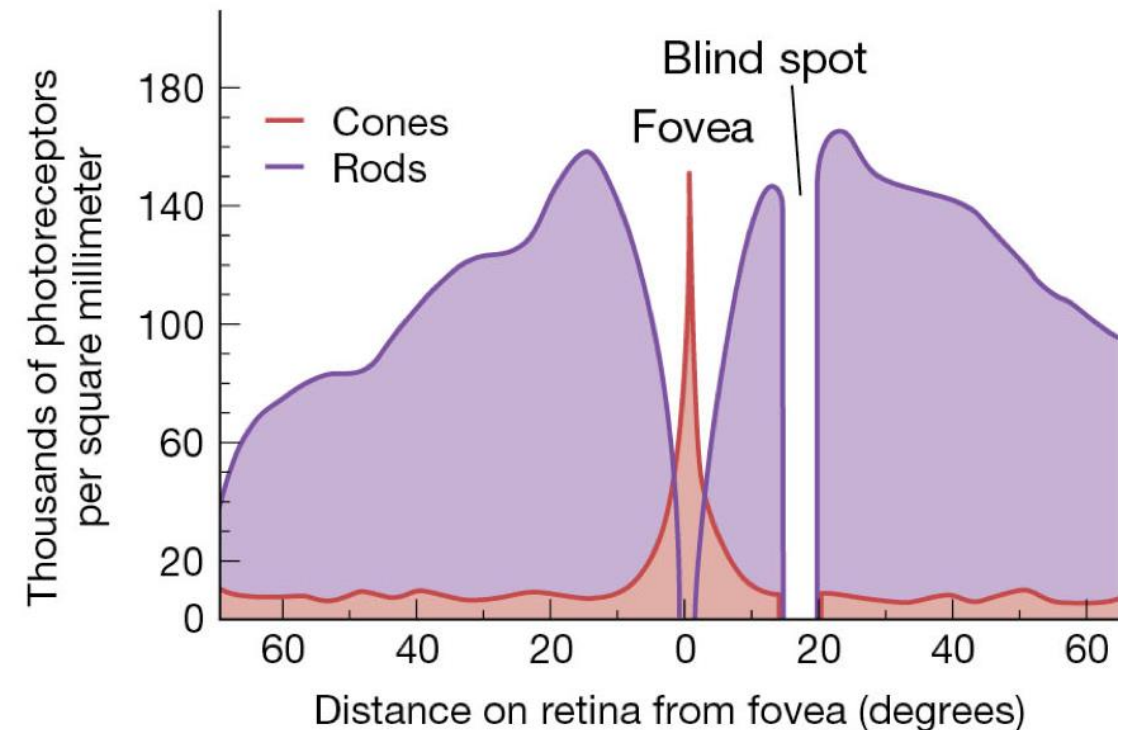
FIGURE 3.1



Copyright © 2019 W. W. Norton & Co., Inc.

Photoreceptors

- Cones
 - 3 kinds
 - Sensitive to different wavelengths
 - Basis of color vision
 - Mostly in the fovea (focus)
 - Not so sensitive – do not work at night
- Rods
 - Only one kind (gray-scale color blind)
 - Mostly in the periphery
 - Sensitive to light (night vision)



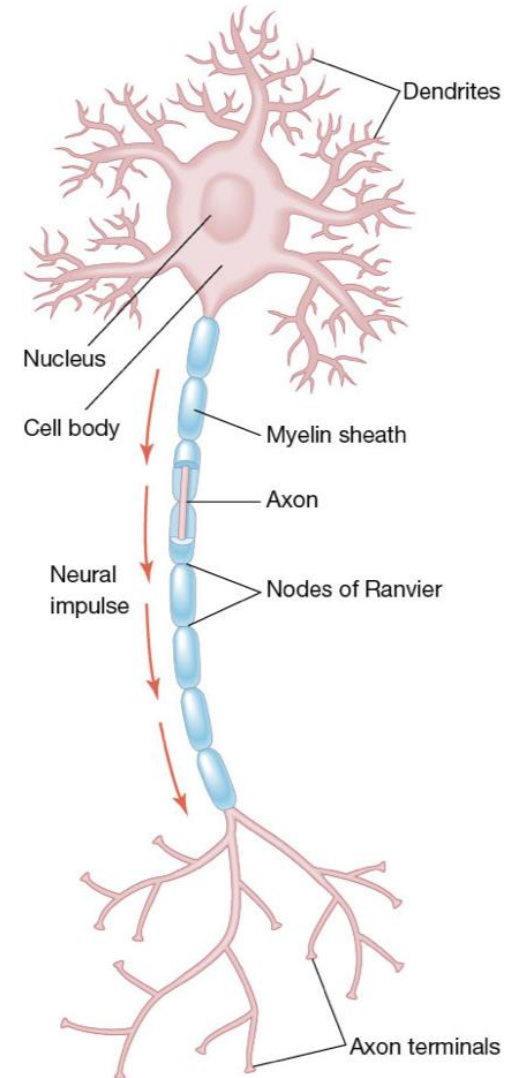
B

Omikron/Science Source
Copyright © 2019 W. W. Norton & Co., Inc.

Neurons code information

- Neurons can be active
 - Action potentials (all-or-none)
 - Rate of action potentials (graded)
 - Level of activation
- Neurons are specific (receptive fields)
 - Cones are specific to location and wavelength
 - Neurons in face area are specific to faces
- Photoreceptors code light
 - Receptive field: where is the light
 - Activation level: how much light is there

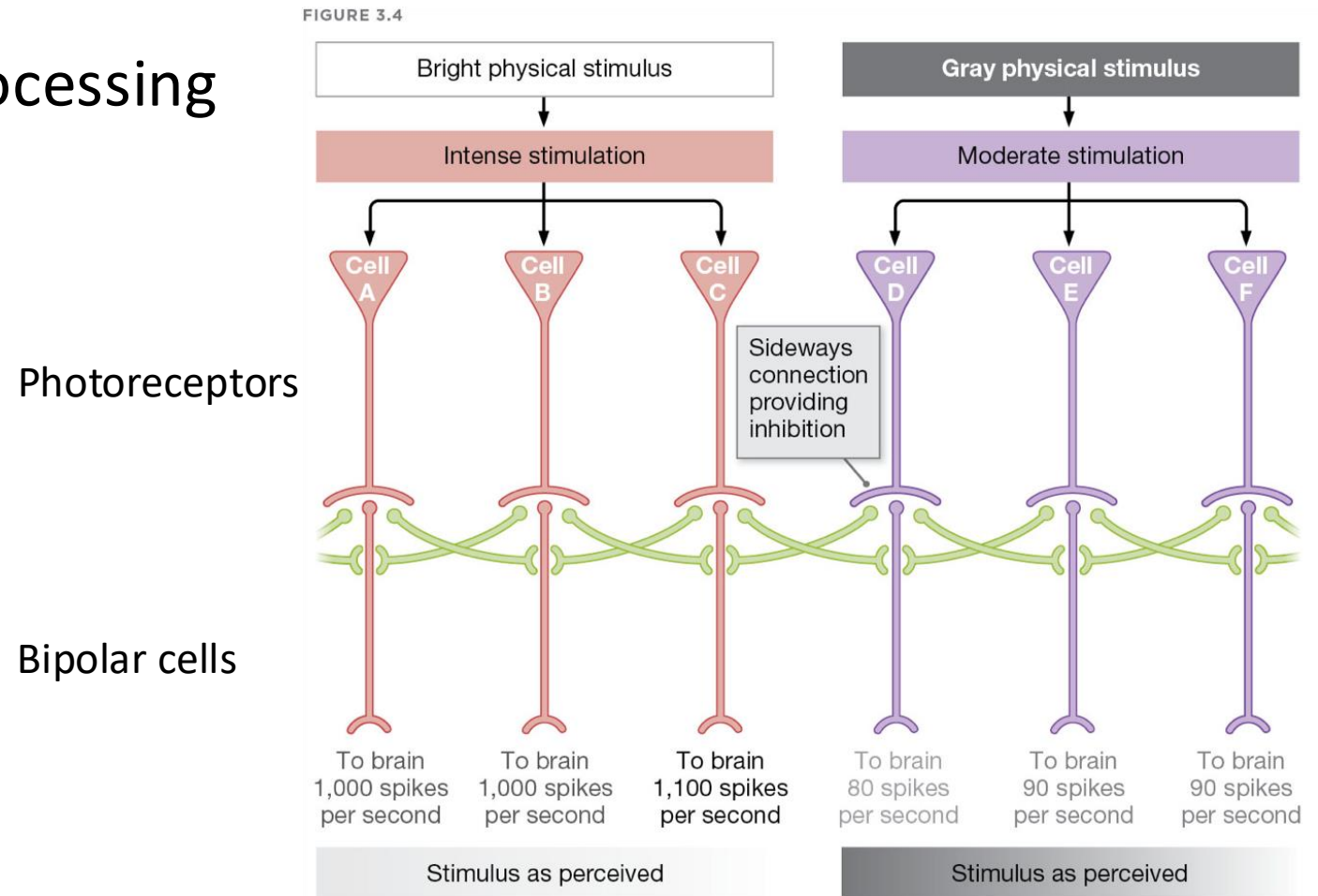
FIGURE 2.12



Copyright © 2019 W. W. Norton & Co., Inc.

Lateral inhibition

- First stage of information processing
- Photo-sensors code light
- Bipolar cells code contrast



Lateral inhibition

1. Assume that we have a 1-dimensional array of M cells. Each cell has an activation level a_m where $m = 2, \dots, M+1$. Each cell receives input j_n from photoreceptor stimulated by light. In addition, each cell receives scaled inhibitory input from neighboring photoreceptors so that

$$a_m = j_m + w(j_{m-1} + j_{m+1}) \quad \text{for } m = 2, \dots, M+1 \quad (3.16)$$

Calculate the activations when input is a 1-dimensional edge, i.e. a steep transition from dark to light. You can use an input vector $\mathbf{j}^T = [0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 1 \ 1 \ 1 \ 1]$ and a weight, $w = -0.5$. Plot the values and explain how you can see that the network enhances contrast.

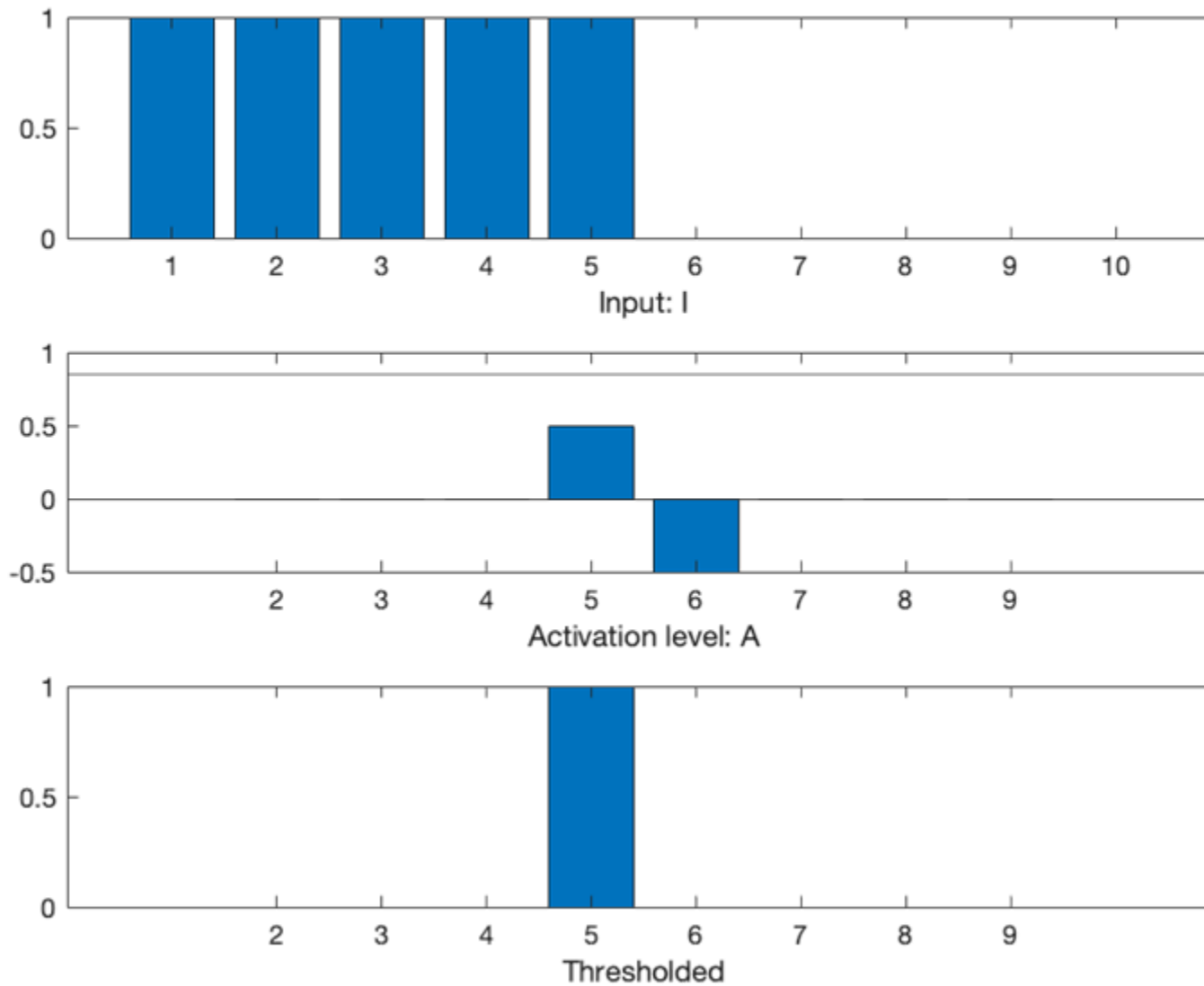
Lateral inhibition

We can express Equation 3.16 as

$$a_m = \sum_{n=-1}^{n=1} j(m+n)k(n) \quad (3.17)$$

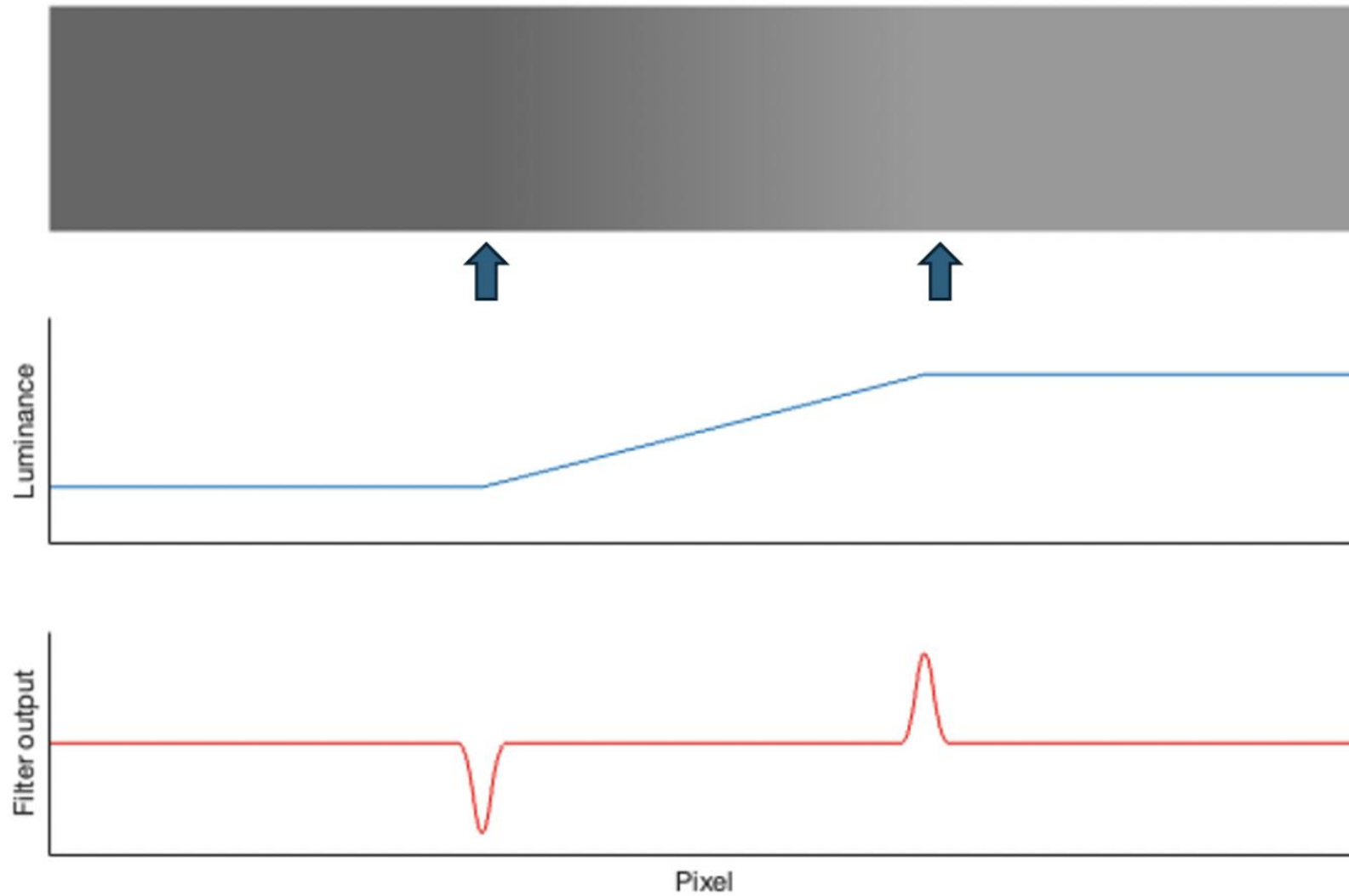
where k is a discrete kernel function with the elements $k(-1) = k(1) = -w$ and $k(0)=1$. Equation 3.17 is the formula of a 1-dimensional convolution showing that the function of the network is to convolve the input with the kernel, k . Recalculate the activation levels using one-dimensional convolution. There are functions for this in most software for scientific computing. Show that you get the same activation values as you got before.

Lateral inhibition



Note that the input is longer than the activations!

Mach Bands



Kernel scales

- How many pixels per photoreceptor?

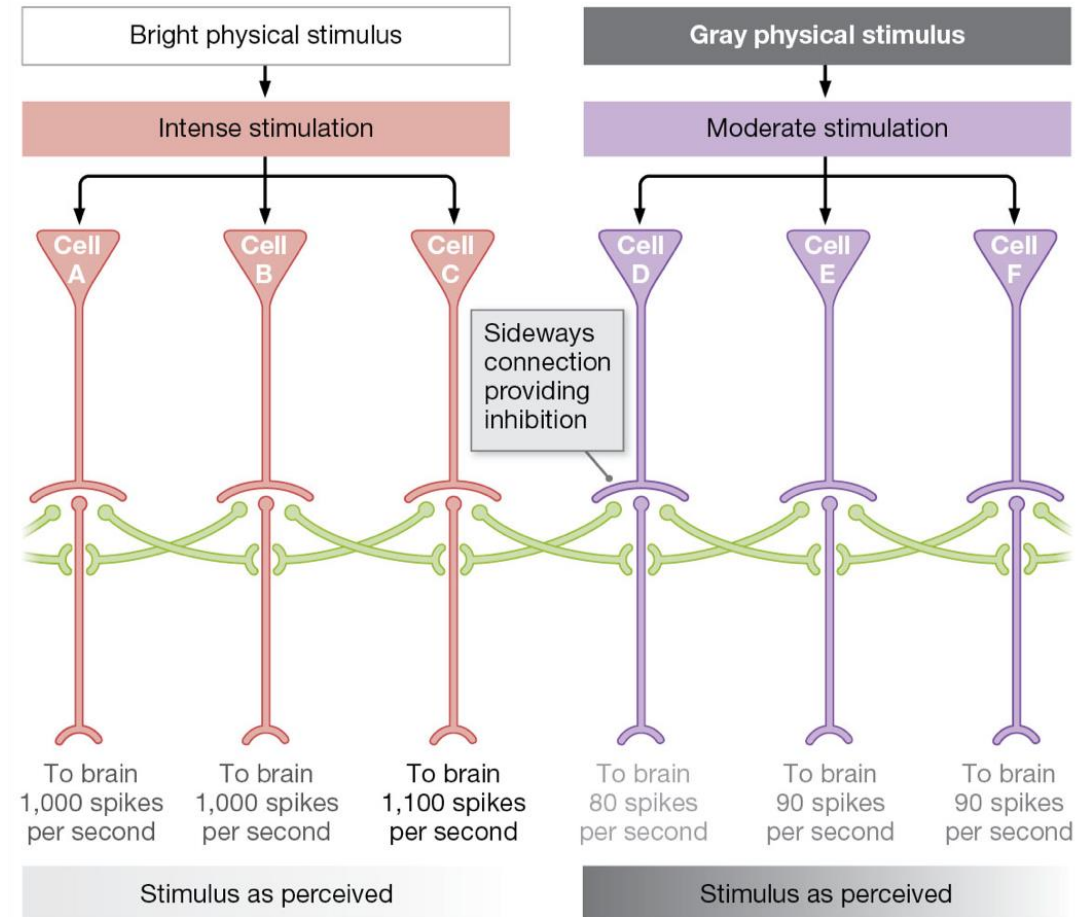
$$\mathbf{k}_1^T = \mathbf{w}^T = [w \quad 1 \quad w]$$

$$\mathbf{k}_2^T = [w \quad w \quad 1 \quad 1 \quad w \quad w]$$

$$\mathbf{k}_3^T = [w \quad w \quad w \quad 1 \quad 1 \quad 1 \quad w \quad w \quad w]$$

- The scale depends on the image!

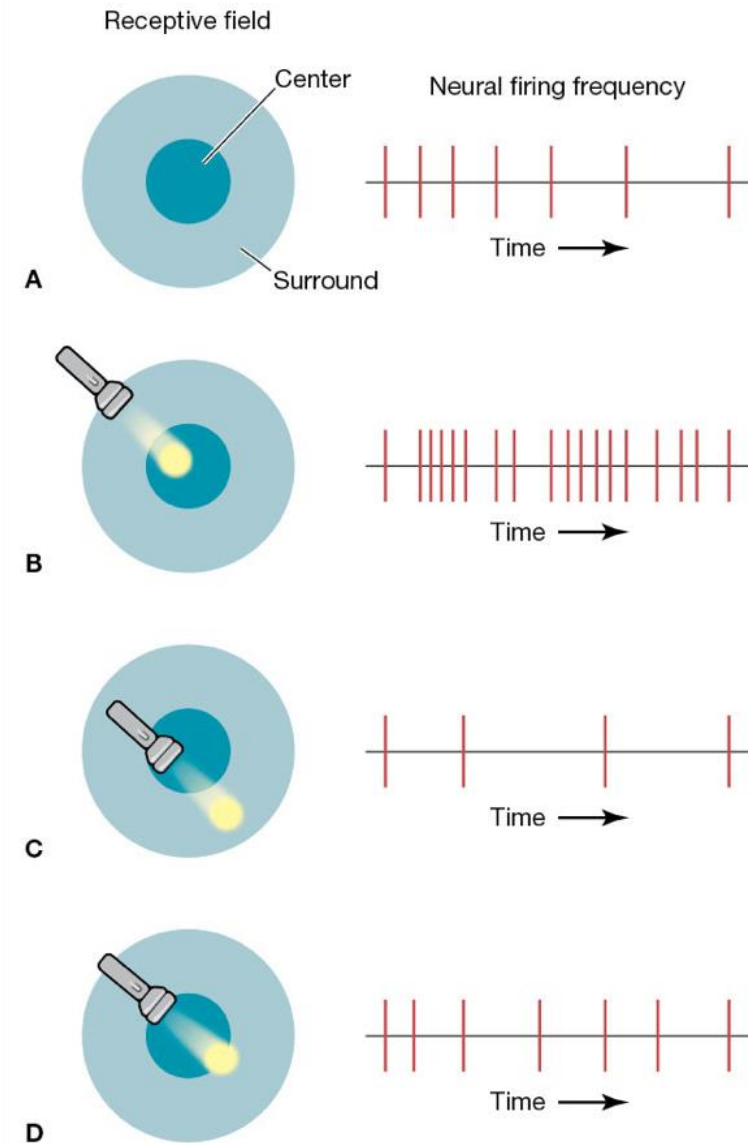
FIGURE 3.4



Receptive fields

- Cells in primary visual cortex code for contrast in 2D
 - Center-surround
 - Receptive field: where is the dot
 - Activation level: how bright is the dot
 - Compared to its surrounding

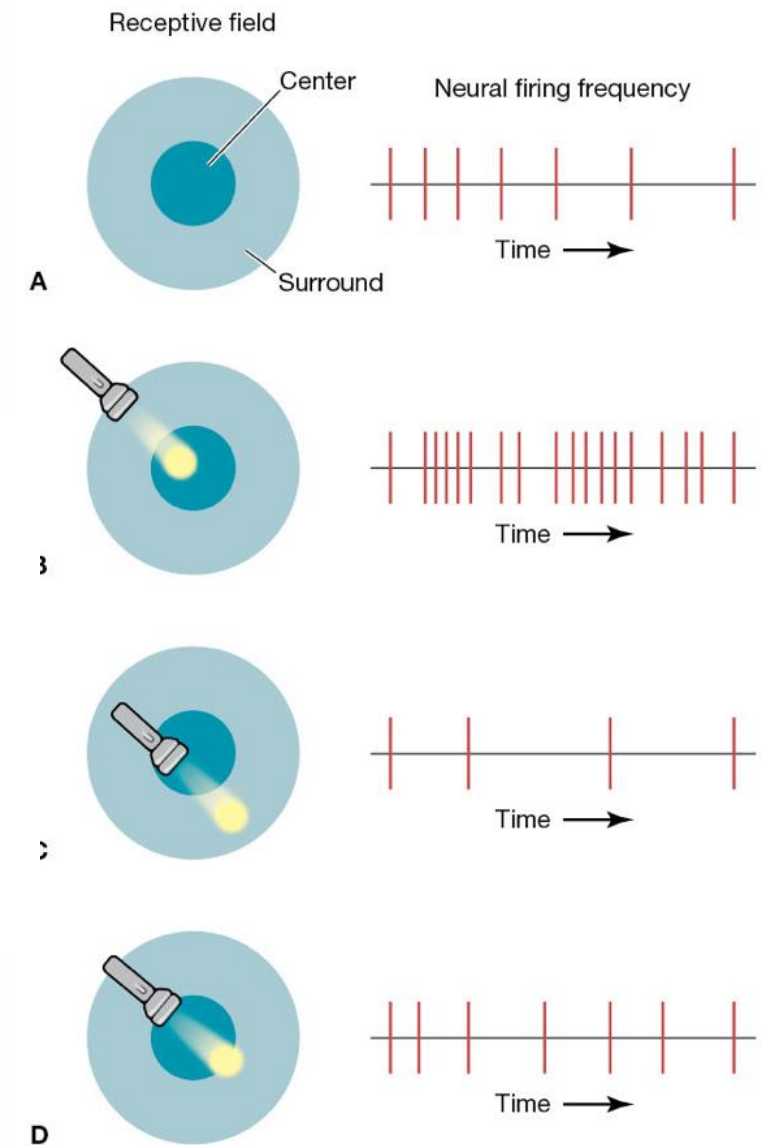
FIGURE 3.6



Mona Lisa – Center-Surround Receptive Fields

$$\mathbf{K}_1 = \begin{bmatrix} w & w & w \\ w & 1 & w \\ w & w & w \end{bmatrix} ; \quad \mathbf{K}_2 = \begin{bmatrix} w & w & w & w & w & w \\ w & w & w & w & w & w \\ w & w & 1 & 1 & w & w \\ w & w & 1 & 1 & w & w \\ w & w & w & w & w & w \\ w & w & w & w & w & w \end{bmatrix}$$

FIGURE 3.6



Mona Lisa

Original



Filtered



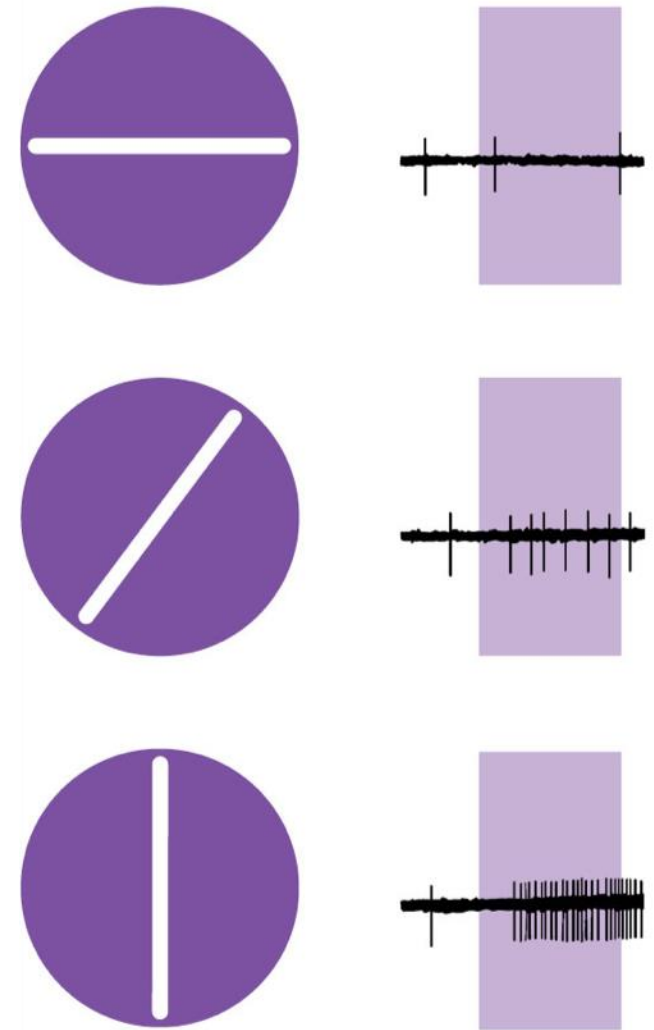
Thresholded



Receptive fields

- Cells in primary visual cortex code for contrast in 2D
 - Edge detectors
 - Receptive field: A localized edge of 90 degree
 - Activation level: How clear/strong is the edge
 - Can be "built" by taking input from multiple center surround cells

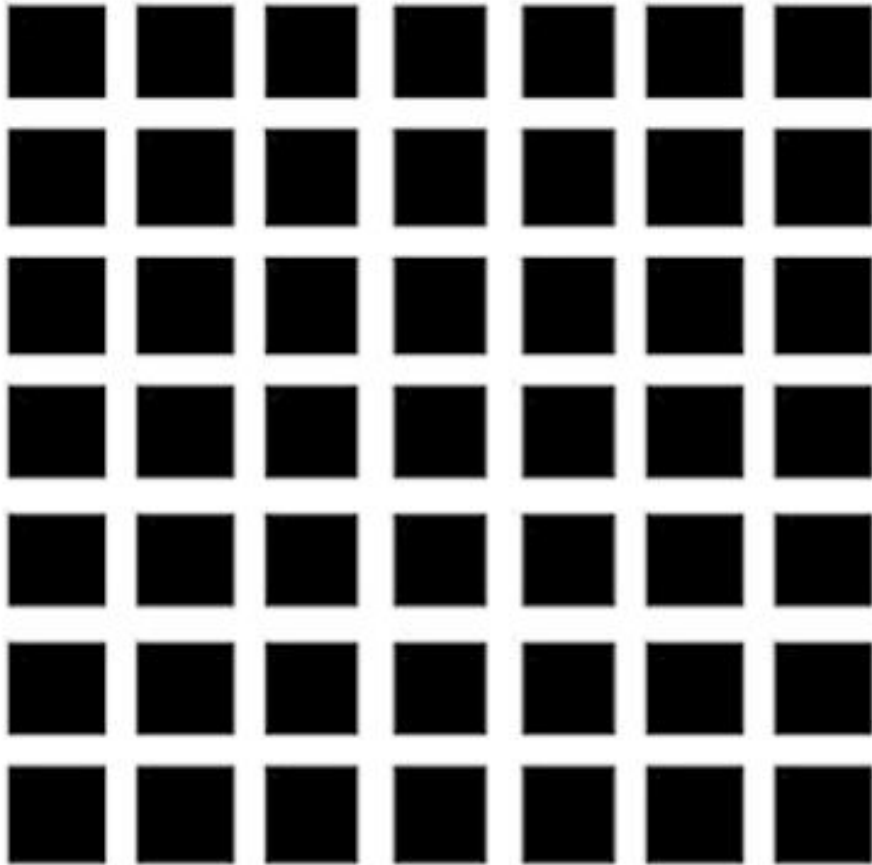
FIGURE 3.7



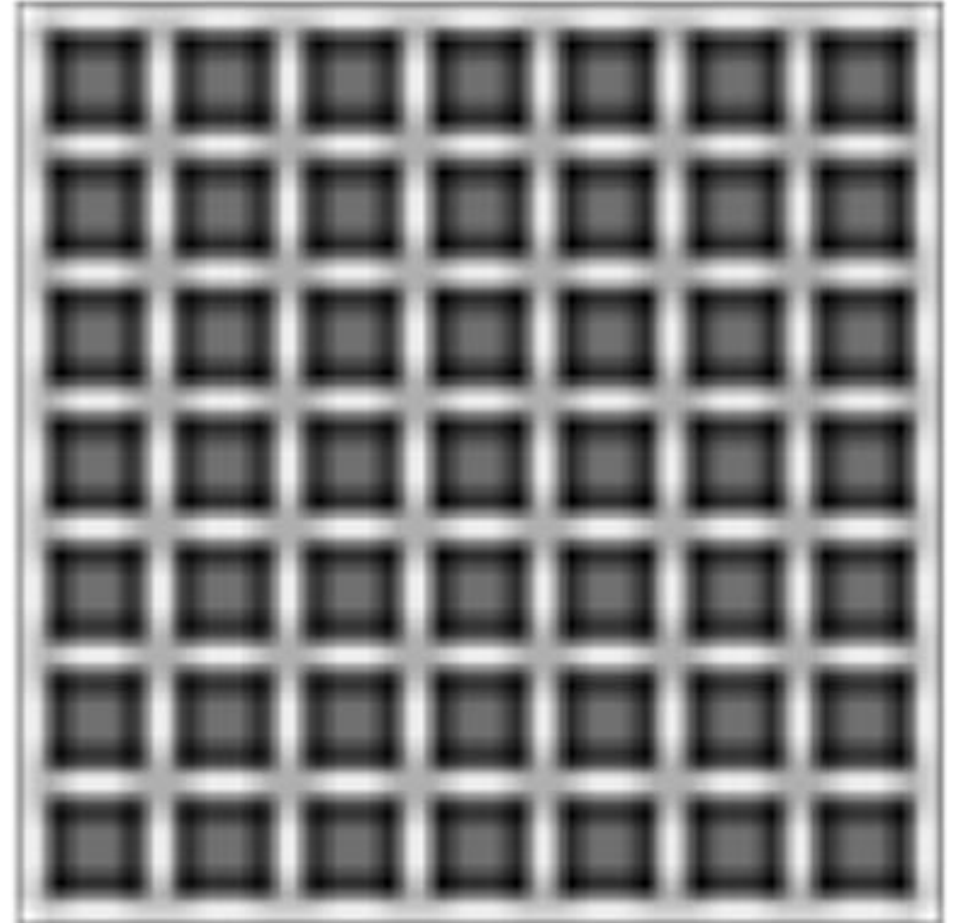
Copyright © 2019 W. W. Norton & Co., Inc.

The Hermann Grid

Original

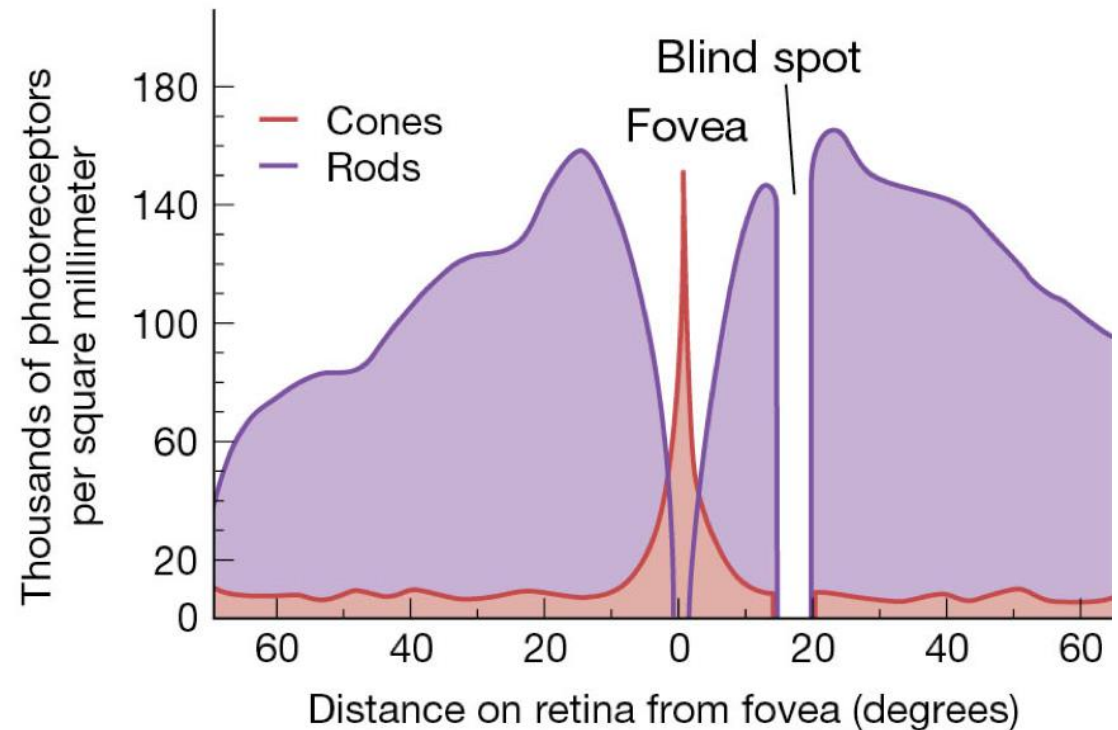


Filtered



The Hermann Grid

- Why does the illusion only occur in peripheral vision?
- Receptive field size is different!



B

Omikron/Science Source
Copyright © 2019 W. W. Norton & Co., Inc.